Gibbs Free Energy, *G* (Revision)

- For a process occurs at constant T & P, with work limited to P-V work,

$$\Delta S_{uni} = \Delta S_{sys} + \Delta S_{surr} = \Delta S_{sys} - \frac{\Delta H_{sys}}{T} > 0$$

for spontaneity

- Multiply the above equation by $-T$, we obtain

$$-T\Delta S_{uni} = -T\Delta S_{sys} + \Delta H_{sys} < 0$$

for spontaneity

- A new thermodynamic function, called the **Gibbs energy** (or **Gibbs free energy**), ***G***, defined as $G = H - TS$, is introduced.

- For a process at constant *T* & *P*,

$$\Delta G = -T\Delta S_{uni} = \Delta H - T\Delta S$$

$$\boldsymbol{\Delta G = \Delta H - T\Delta S}$$

Gibbs Free Energy, *G* (Revision)

$$\Delta G = \Delta H - T\Delta S$$

- All the terms refer to changes for the **system**.
- As $\Delta G = -T\Delta S_{uni}$, ΔG is negative when ΔS_{uni} is positive.

$\Delta G = \Delta H - T\Delta S < 0$, the process is spontaneous.

$\Delta G = \Delta H - T\Delta S > 0$, the process is non-spontaneous.

$\Delta G = \Delta H - T\Delta S = 0$, the process is reversible and the system has reached equilibrium.

Applying the Gibbs Energy Criteria for Spontaneous Change

Case	ΔH	ΔS	ΔG	Result	Example
1.	–	+	–	spontaneous at all temp.	$2\,N_2O(g) \longrightarrow 2\,N_2(g) + O_2(g)$
2.	–	–	–	spontaneous at low temp.	$H_2O(l) \longrightarrow H_2O(s)$
			+	nonspontaneous at high temp.	
3.	+	+	+	nonspontaneous at low temp.	$2\,NH_3(g) \longrightarrow N_2(g) + 3\,H_2(g)$
			–	spontaneous at high temp.	
4.	+	–	+	nonspontaneous at all temp.	$3\,O_2(g) \longrightarrow 2\,O_3(g)$

Class Practice 6

Which of the four cases would apply to each of the following reactions?

(a) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $\Delta_r H^{\ominus} = -92.22$ kJ/mol

(b) $2C(\text{graphite}) + 2H_2(g) \rightarrow C_2H_4(g)$, $\Delta_r H^{\ominus} = 52.26$ kJ/mol

Case	ΔH	ΔS	ΔG
1.	−	+	−
2.	−	−	− or +
3.	+	+	+ or −
4.	+	−	+

Gibbs Free Energy Change of a System of Variable Composition $\Delta_r G^{\ominus}$

- ΔG is the key quantity for predicting the spontaneity of change under the condition of constant T and constant P.
- In such conditions, the change of Gibbs energy arises from changes in the amounts of substances (i.e. from a change in composition of the system)
- The ***standard Gibbs energy of reaction***, $\Delta_r G°$, is the Gibbs energy change **per mole of reaction** for the following process.

Pure, unmixed reactants (each in its standard state) ⟶ Pure, unmixed products (each in its standard state)

The Standard State of Substances:

State of Matter	Standard state
Solid	Pure solid
Liquid	Pure liquid
Gas	1 bar
Solution	1 M

Calculation of $\Delta_r G^{\ominus}$ based on $\Delta_f G^{\ominus}$

$$a\mathrm{A} + b\mathrm{B} + \ldots \longrightarrow c\mathrm{C} + d\mathrm{D} + \ldots$$

The unit of $\Delta_r G^{\ominus}$ is J mol^{-1}

$$\Delta_r G^\circ = \underbrace{[c\,\Delta_f G^\circ_C + d\,\Delta_f G^\circ_D + \ldots]}_{\text{weighted sum of } \Delta_f G^\circ \text{ values for the products}} - \underbrace{[a\,\Delta_f G^\circ_A + b\,\Delta_f G^\circ_B + \ldots]}_{\text{weighted sum of } \Delta_f G^\circ \text{ values for the reactants}}$$

- $\Delta_f G^{\ominus}$, the **standard Gibbs energy of formation**, is the Gibbs energy change per mole for a reaction in which a substance in its standard state is formed from its elements in their reference forms in their standard states.
- $\Delta_f G^{\ominus} = 0$ for an element in its reference form at a pressure of 1 bar.
- $\Delta_r G^{\ominus}$ can be calculated using

(i) $\Delta_r G^\circ = [c\,\Delta_f G^\circ_C + d\,\Delta_f G^\circ_D + \ldots] - [a\,\Delta_f G^\circ_A + b\,\Delta_f G^\circ_B + \ldots]$, or

(ii) $\Delta G = \Delta H - T\Delta S$.

Class Practice 7

Calculation of $\Delta_r G^{\ominus}$ based on $\Delta_f G^{\ominus}$

Determine $\Delta_r G^{\ominus}$ for the complete combustion of benzene at 25 °C using the data given below. Hence comment on the spontaneity of the following reaction at this temperature.

Thermodynamic data at 25 °C

	$\Delta_f G^{\ominus}$ (kJ/mol)
C_6H_6 (l)	–124.5
CO_2 (g)	–394.4
H_2O (l)	–237.2

Summary: Thermodynamic Quantities

$$\Delta G^{\ominus} = \Delta H^{\ominus} - T\Delta S^{\ominus}$$

Term	Name	Explanation
ΔG°	Gibbs free-energy change	The energy difference between reactants and products. When ΔG° is negative, the reaction is **exergonic**, has a favorable equilibrium constant, and can occur spontaneously. When ΔG° is positive, the reaction is **endergonic**, has a unfavorable equilibrium constant, and cannot occur spontaneously.
ΔH°	Enthalpy change	The heat of reaction, or difference in strength between the bonds broken in a reaction and the bonds formed. When ΔH° is negative, the reaction releases heat and is **exothermic**. When ΔH° is positive, the reaction absorbs heat and is **endothermic**.
ΔS°	Entropy change	The change in molecular randomness during a reaction. When ΔS° is negative, randomness decreases. When ΔS° is positive, randomness increases.

Gibbs Free Energy of Reaction for Non-standard Conditions, $\Delta_r G$

- Consider a system containing N_2 (g), H_2 (g), and NH_3 (g), treated as ideal gases, and let n_{N2}, n_{H2} , n_{NH3} represent the initial amounts, in moles, of each gas. Suppose the composition of the system changes, at constant T and P, because the following reaction proceeds by an infinitesimal amount, $d\xi$.

(NOTE: ξ is the extent of reaction)

$$N_2(g) + 3\,H_2(g) \underset{d\xi<0}{\overset{d\xi>0}{\rightleftharpoons}} 2\,NH_3(g)$$

	$N_2(g)$	$3\,H_2(g)$	$2\,NH_3(g)$	
Initial state:	n_{N_2}	n_{H_2}	n_{NH_3}	G_i
Change:	$-d\xi$	$-3\,d\xi$	$+2\,d\xi$	$+dG$
Final state:	$n_{N_2} - d\xi$	$n_{H_2} - 3\,d\xi$	$n_{NH_3} + 2\,d\xi$	$G_f = G_i + dG$

- The sign of $d\xi$ determines the direction of reaction:
  - a positive value ($d\xi > 0$) describes reaction from left to right;
  - a negative value ($d\xi < 0$) describes reaction from right to left.

Gibbs Free Energy of Reaction for Non-standard Conditions, $\Delta_r G$

- To decide to which direction the reaction is spontaneous, we need to know how G changes with ξ, the extent of reaction, and more specifically, **the rate of change of G with respect to ξ**.

- The rate of change of G with respect to the extent of reaction is represented by the symbol $\Delta_r G$, called the **Gibbs energy of reaction**.

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$$

- The quantity Q is called the **reaction quotient**.

- Since the above equation relates the value of $\Delta_r G$, for arbitrary non-standard conditions to its value under standard conditions, the term $RT\ln Q$ includes the effects of bringing the system from standard conditions (pure, unmixed substances, each at $P^{\ominus} = 1$ bar) to the actual non-standard conditions (a mixture of substances with a total pressure P).

Gibbs Free Energy of Reaction for Non-standard Conditions, $\Delta_r G$

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$$

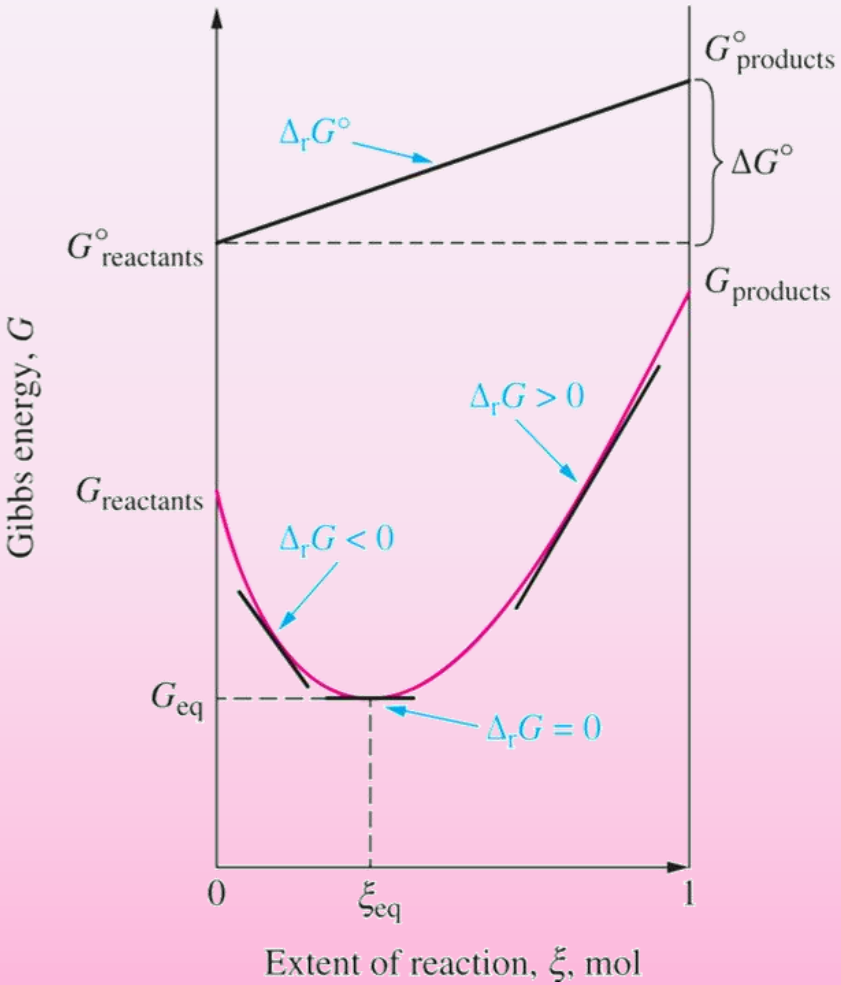
- $\Delta_r G$ is the slope of the tangent at a particular point in a graph of G versus ξ.

1. If $\Delta_r G < 0$, then G decreases as ξ increases.
2. If $\Delta_r G > 0$, then G increases as ξ increases.
3. If $\Delta_r G = 0$, then the system has attained the minimum possible value of G.

- Predicting the direction of spontaneous chemical change:

$\Delta_r G < 0$ Left to right ($\longrightarrow$)
$\Delta_r G > 0$ Right to left ($\longleftarrow$)
$\Delta_r G = 0$ Equilibrium ($\rightleftharpoons$)

Variation of G at constant T and constant P



Summary: $\Delta_r G$ and $\Delta_r G^{\ominus}$

- $\Delta_r G^{\ominus}$ is often calculated using thermochemical data for pure substances in their standard states.

$$\Delta G^{\ominus} = \Delta H^{\ominus} - T\Delta S^{\ominus}$$

- $\Delta_r G$ represents the rate of change of Gibbs free energy with respect to the extent of reaction (ξ) for a system under the specified conditions of composition. Its value can be calculated when the values of $\Delta_r G^{\ominus}$ and Q are determined. The sign of $\Delta_r G$ tells the direction of spontaneous change.

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$$

- Q represents the reaction quotient, and its value depends on the composition of the system.

The Thermodynamic Reaction Quotient, *Q*

- Consider the general equation below,

$$aA + bB + \ldots \longrightarrow cC + dD + \ldots$$

- the **reaction quotient, *Q*,** is given by the expression,

$$Q = \frac{a_C^c \times a_D^d \times \cdots}{a_A^a \times a_B^b \times \cdots}$$

(where *a* represents the **activity** of the substances)

Example: $N_2(g) + 3\,H_2(g) \longrightarrow 2\,NH_3(g)$

$$Q = \frac{(a_{NH_3})^2}{(a_{N_2})(a_{H_2})^3}$$

$$Q = \frac{(P_{NH_3}/P^\circ)^2}{(P_{N_2}/P^\circ)(P_{H_2}/P^\circ)^3}$$

Activities of Solids, Liquids, Gases, and Solutes*

Form of Substance	Activity
X(s)	$a = 1$
X(l)	$a = 1$
X(g)	$a = P_X/P^\circ$
X(aq)	$a = [X]/c^\circ$

*In these relationships, P° = 1 bar and c° = 1 mol L^{-1}.

Example 3

Calculating $\Delta_r G$ from Thermodynamic Data and Specified Reaction Conditions

A system contains H_2, N_2, and NH_3 gases, each with a partial pressure of 0.100 bar. The temperature is held constant at 298.15 K. Calculate the Gibbs energy of reaction, $\Delta_r G$, for the formation of $NH_3(g)$ from $N_2(g)$ and $H_2(g)$ under these conditions and predict whether formation or consumption of NH_3 is spontaneous. Given the $\Delta_f G^{\ominus}$ [NH_3 (g)] = –16.45

Suggested Solution:

$$\Delta_r G^\circ = 2\Delta_f G^\circ[NH_3(g)] - \Delta_f G^\circ[N_2(g)] - 3\Delta_f G^\circ[H_2(g)]$$
$$= [2(-16.45) - 1(0) - 3(0)]\ \text{kJ mol}^{-1}$$
$$= -32.90\ \text{kJ mol}^{-1}$$

$$Q = \frac{(P_{NH_3}/P^\circ)^2}{(P_{N_2}/P^\circ)(P_{H_2}/P^\circ)^3} = \frac{(0.10)^2}{(0.10)(0.10)^3} = 100$$

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$$

$$\Delta_r G = \Delta_r G^\circ + RT \ln Q$$
$$= -32.90\ \text{kJ mol}^{-1} + (8.314 \times 10^{-3}\ \text{kJ mol}^{-1}\ \text{K}^{-1})(298.15\ \text{K}) \ln 100$$
$$= -21.5\ \text{kJ mol}^{-1}$$

As the rate of change of Gibbs free energy with respect to the extent of reaction (ξ) is negative, for the given initial conditions, the reaction proceeds spontaneously from left to right, leading to the formation of more NH_3.

Relationship of $\Delta_r G^{\ominus}$ to the Equilibrium Constant K

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q$$

- At equilibrium, $\Delta_r G = 0$ and $Q = K$. The above equation can be written as:

$$\Delta_r G^{\ominus} = -RT \ln K$$

- Combine both equations,

$$\Delta_r G = \Delta_r G^{\ominus} + RT \ln Q = -RT \ln K + RT \ln Q$$

$$\Delta_r G = RT \ln (Q/K)$$

- Using Q and K to Predict the Direction of Spontaneous Chemical Change

Condition	Spontaneous Reaction
$Q < K$	Left to right ($\longrightarrow$)
$Q > K$	Right to left ($\longleftarrow$)
$Q = K$	Equilibrium ($\rightleftharpoons$)

The Equilibrium Constant, *K*

- Consider the general equation for a reaction that has reached **equilibrium**.

$$a\,\mathrm{A} + b\,\mathrm{B} + \ldots \rightleftharpoons c\,\mathrm{C} + d\,\mathrm{D} + \ldots$$

- The **equilibrium constant, *K*,** is given by the expression,

$$K = \frac{a_C^c \times a_D^d \times \cdots}{a_A^a \times a_B^b \times \cdots}$$

(Note that for *K*, **equilibrium** values of the activities are implied.)

- No matter the starting composition of reactants and products, the same ratio of concentrations is achieved at equilibrium under constant temperature.

Scenario 1: Reactions Involving Only Gases

$$K = \frac{(P_{C,eq}/P^\circ)^c (P_{D,eq}/P^\circ)^d \ldots}{(P_{A,eq}/P^\circ)^a (P_{B,eq}/P^\circ)^b \ldots} = \frac{(P_{C,eq})^c (P_{D,eq})^d \ldots}{(P_{A,eq})^a (P_{B,eq})^b \ldots} \times \left(\frac{1}{P^\circ}\right)^{\Delta\nu}$$

Note that:

$\Delta\nu = (c + d + \cdots) - (a + b + \cdots)$

$P^\circ = 1$ bar

$$K_P = \frac{P_{C,eq}^c P_{D,eq}^d \cdots}{P_{A,eq}^a P_{B,eq}^b \cdots}$$

$$K = K_p \times (1/P^\circ)^{\Delta\nu}$$

The Equilibrium Constant, *K*

Scenario 2: Reactions in Aqueous Solution

❖ The activity of A (aq) is expressed as $a_A = [A]/c^\circ$.

$$K = K_c \times (1/c^\circ)^{\Delta\nu}$$

where $K_c = \dfrac{[C]_{eq}^{c}\,[D]_{eq}^{d}\cdots}{[A]_{eq}^{a}\,[B]_{eq}^{b}\cdots}$ and $\Delta\nu = (c + d + \cdots) - (a + b + \cdots)$

Scenario 3: Reactions in a Heterogeneous System

Example: $2\,Al(s) + 6\,H^+(aq) \rightleftharpoons 2\,Al^{3+}(aq) + 3\,H_2(g)$

$$K = \frac{\left(a_{Al^{3+},eq}\right)^2\left(a_{H_2,eq}\right)^3}{\left(a_{Al,eq}\right)^2\left(a_{H^+,eq}\right)^6} \approx \frac{\left([Al^{3+}]_{eq}/c^\circ\right)^2\left(P_{H_2,eq}/P^\circ\right)^3}{(1)^2\left([H^+]_{eq}/c^\circ\right)^6}$$

$$= \frac{[Al^{3+}]_{eq}^2\,P_{H_2,eq}^3}{[H^+]_{eq}^6} \times (c^\circ)^4 \times \left(\frac{1}{P^\circ}\right)^3$$

Class Practice 8

Writing Thermodynamic Equilibrium Constant Expressions

For the following reversible reactions, write thermodynamic equilibrium constant expressions, making appropriate substitutions for activities. Then relate K to K_c or K_p, where this can be done.

(a) The water gas reaction

$$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$$

(b) Formation of a saturated aqueous solution of lead(II) iodide, a very slightly soluble solute

$$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2\,I^-(aq)$$

(c) Oxidation of sulphide ion by oxygen gas (used in removing sulphides from wastewater, as in pulp and paper mills)

$$O_2(g) + 2\,S^{2-}(aq) + 2\,H_2O(l) \rightleftharpoons 4\,OH^-(aq) + 2\,S(s)$$



Class Practice 8 (Continued)

Class Practice 9

Calculating the Equilibrium Constant of a Reaction from the Standard Gibbs Free Energy of Reaction

Determine the equilibrium constants K and K_c at 298.15 K for the dissolution of magnesium hydroxide in an acidic solution.

$$Mg(OH)_2(s) + 2\,H^+(aq) \rightleftharpoons Mg^{2+}(aq) + 2\,H_2O(l)$$

Substance	$Mg(OH)_2$ (s)	H^+ (aq)	Mg^{2+} (aq)	H_2O (l)
$\Delta_f G°$ (kJ/mol)	–833.5	0	–454.8	–237.1

Interpreting the Values of $\Delta_r G^{\ominus}$ and K

- From $\Delta_r G^{\ominus} = -RT \ln K$, we can obtain the following.

$$K = e^{\frac{-\Delta_r G^{\ominus}}{RT}}$$

- This expression can be used to calculate values of K at 298 K for different values of $\Delta_r G^{\ominus}$.

Significance of the magnitude of $\Delta_r G°$ and K at 298 K

$\Delta_r G°$, kJ mol^{-1}	K	Significance
+200	9.1×10^{-36}	Equilibrium favors reactants
+100	3.0×10^{-18}	
+50	1.7×10^{-9}	
+10	1.8×10^{-2}	
+1.0	6.7×10^{-1}	
0	1.0	Equilibrium calculation is necessary
−1.0	1.5	
−10	5.6×10^{1}	
−50	5.8×10^{8}	
−100	3.3×10^{17}	Equilibrium favors products
−200	1.1×10^{35}	

Interpreting the Values of $\Delta_r G^{\ominus}$ and K

Consider the following gas-phase reaction,

$$a\,\mathrm{A(g)} + b\,\mathrm{B(g)} + \ldots \rightleftharpoons c\,\mathrm{C(g)} + d\,\mathrm{D(g)} + \ldots$$

$$K = \frac{(P_{C,eq})^c (P_{D,eq})^d \ldots}{(P_{A,eq})^a (P_{B,eq})^b \ldots} \times \left(\frac{1}{P^{\square}}\right)^{\Delta\upsilon}$$

❖ For ξ = 0 mol, the system contains stoichiometric amounts of reactants (*a* mol A and *b* mol B) at a total pressure *P*. For ξ = 1 mol, the system contains stoichiometric amounts of products (*c* mol C and *d* mol D) at a total pressure P.

❖ Plots of ***G*** **vs** extent of reaction **(*ξ*)** for three different situations:

(a) an intermediate value of *K*,

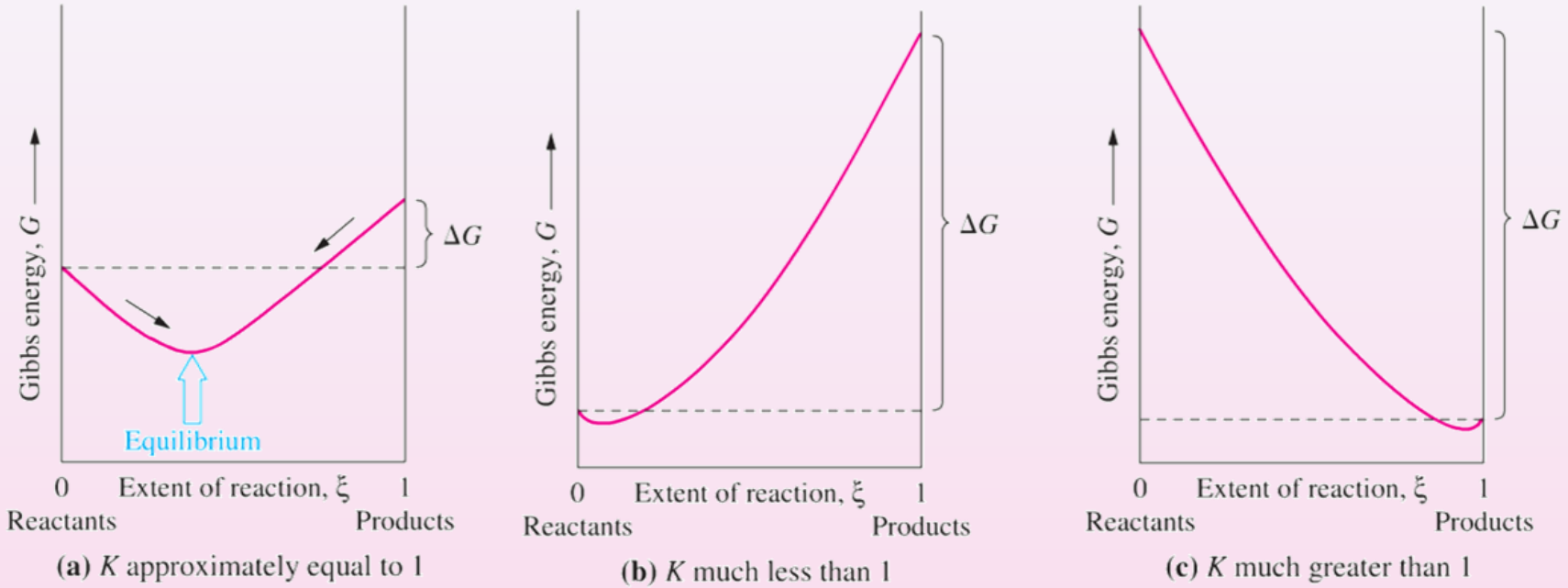
(b) a very small value of *K*, and

(c) a very large value of *K*.

Interpreting the Values of $\Delta_r G^{\ominus}$ and K

The position of equilibrium for small, large, and intermediate values of K

$$a\,\mathrm{A} + b\,\mathrm{B} \rightarrow c\,\mathrm{C} + d\,\mathrm{D}$$



(a) When $K \sim 1$, the value of ΔG is small and the equilibrium position corresponds to having appreciable amounts of both reactants and products.

(b) When $K \ll 1$, ΔG has a large positive value and the equilibrium position corresponds to an equilibrium position closer to reactants (a small value of ξ).

(c) When $K \gg 1$, ΔG has a large negative value and the equilibrium position corresponds to an equilibrium position closer to products (a larger value of ξ).

Interpreting the Values of $\Delta_r G^{\ominus}$ and K

Summary:

- When $K \sim 1$, the equilibrium position corresponds to having appreciable amounts of both reactants and products. Complete conversion of reactants to products would be accompanied by a **relatively small change** in G.

- A **small K ($K << 1$)** value indicates that products in their standard states are **thermodynamically less stable** than reactants in their standard states and the equilibrium position is **closer to reactants**. Complete conversion of reactants to products would be accompanied by a **large increase** in G.

- A **large K ($K >> 1$)** value indicates that products in their standard states are **thermodynamically more stable** than reactants in their standard states and the equilibrium position is **closer to products**. Complete conversion of reactants to products would be accompanied by a **large decrease** in G.

$\Delta_r G^{\ominus}$ and K as Functions of Temperature

- Use $\Delta G^{\ominus} = \Delta H^{\ominus} - T\Delta S^{\ominus}$ and $\Delta G^{\ominus} = -RT \ln K$, and assume $\Delta H^{\ominus}$ and $\Delta S^{\ominus}$ are independent of temperature,

For T_1, $$\ln K_1 = \frac{-\Delta G^{\ominus}}{RT_1} = \frac{-\Delta H^{\ominus}}{RT_1} + \frac{\Delta S^{\ominus}}{R} \quad \ldots (1)$$

For T_2, $$\ln K_2 = \frac{-\Delta G^{\ominus}}{RT_2} = \frac{-\Delta H^{\ominus}}{RT_2} + \frac{\Delta S^{\ominus}}{R} \quad \ldots (2)$$

- Combining the equation, (2) – (1),

$$\ln \frac{K_2}{K_1} = -\frac{\Delta H^{\circ}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

the van't Hoff equation

where $R = 8.3145$ J mol^{-1} K^{-1}

- The equation is often used to calculate an equilibrium constant K_2 at T_2, when K_1 at T_1 and ΔH° are known.



Class Practice 10

Determining the Relationship Between an Equilibrium Constant and Temperature by Using Equations for Gibbs Energy of Reaction

At what temperature will the equilibrium constant for the formation of NOCl (g) be $K = 1.00 \times 10^3$?

$$2\,NO(g) + Cl_2(g) \rightleftharpoons 2\,NOCl(g)$$

Data for this reaction at 25 °C are

$$\Delta_r H = -77.1\ \text{kJ mol}^{-1} \quad \Delta_r S^\circ = -121.3\ \text{J mol}^{-1}\,\text{K}^{-1}$$

Class Practice 11

Relating Equilibrium Constants and Temperature Through the van't Hoff Equation

Use the data below to estimate the temperature at which $K = 1.00 \times 10^6$ for the reaction

$$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$$

(Given the equilibrium constants K for the reaction at several temperatures on the right.)

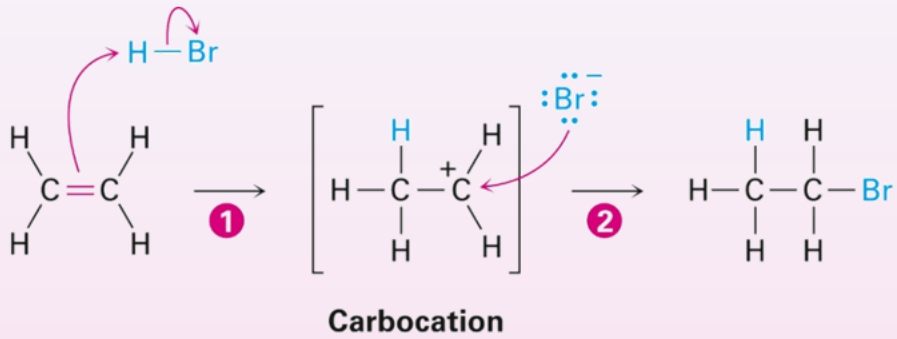
T, K	*K*
800	9.1×10^2
850	1.7×10^2
900	4.2×10^1
950	1.0×10^1
1000	3.2×10^0
1050	1.0×10^0
1100	3.9×10^{-1}
1170	1.2×10^{-1}

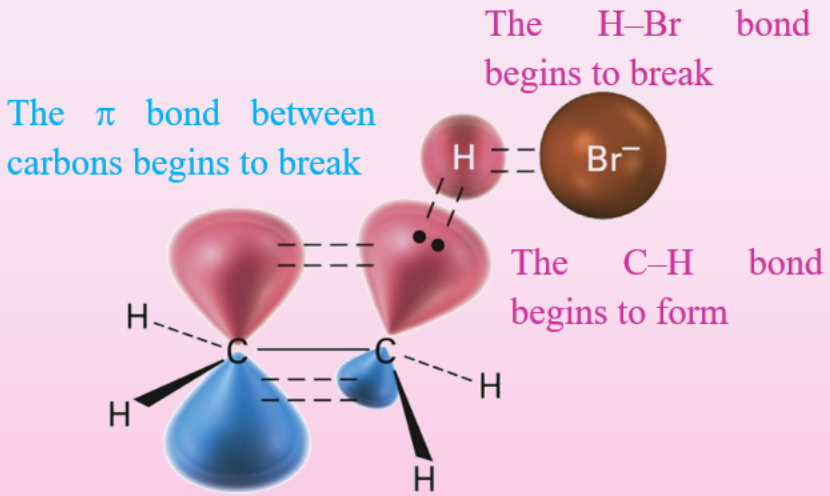


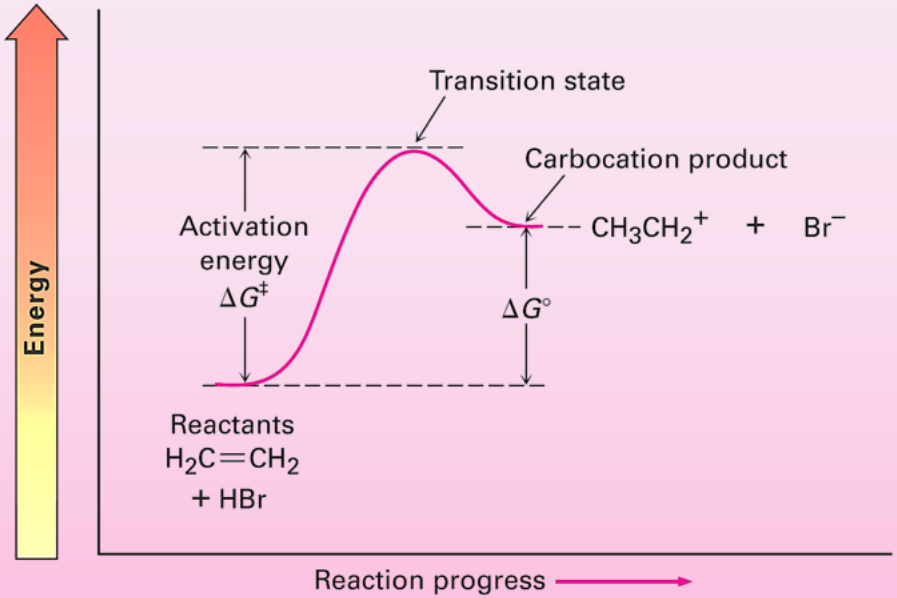
Class Practice 11 (Continued)

Describing a Reaction: Energy Profiles and Transition States (Revision)

- Consider hydrogenhalogenation of ethylene, the electrophilic addition reaction
- The highest energy point in a reaction step is called the **transition state.**



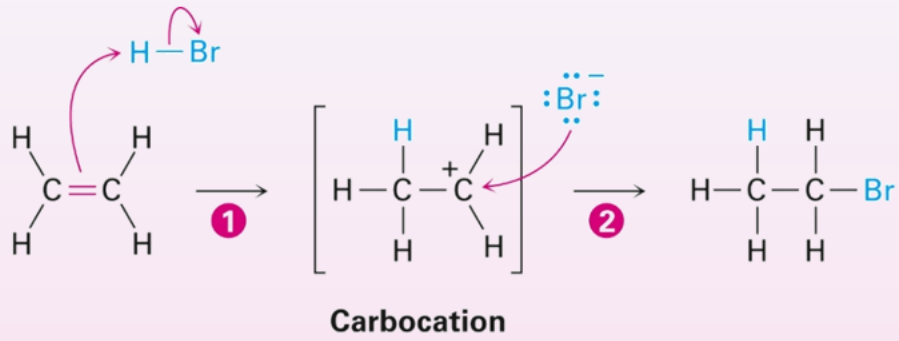


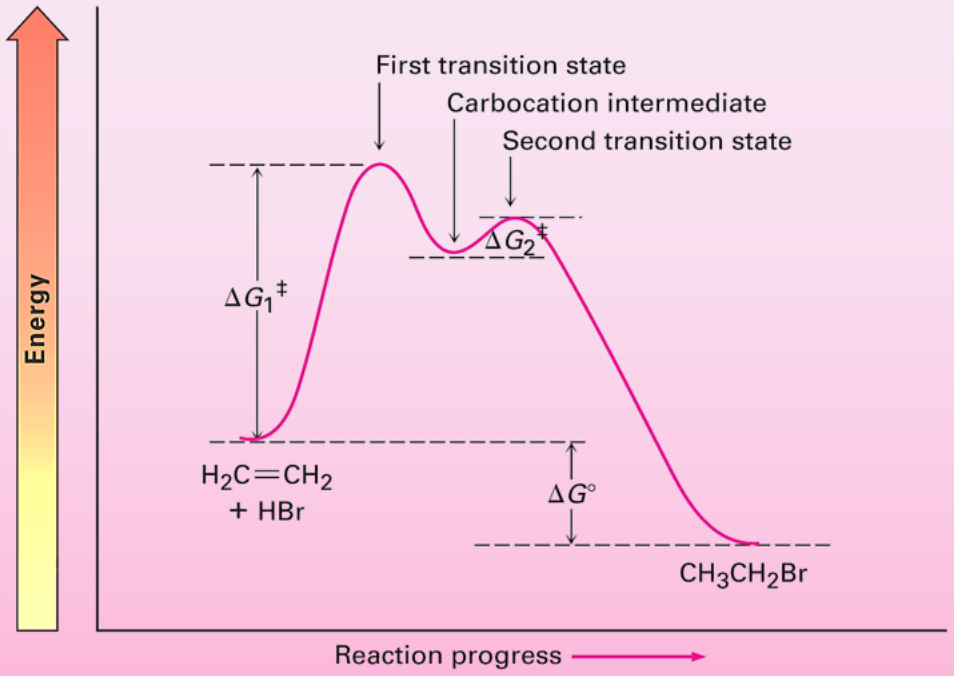


- The energy needed to go from reactant to transition state is the **activation energy** ($\Delta G^{\ddagger}$)

Describing a Reaction: Intermediate (Revision)

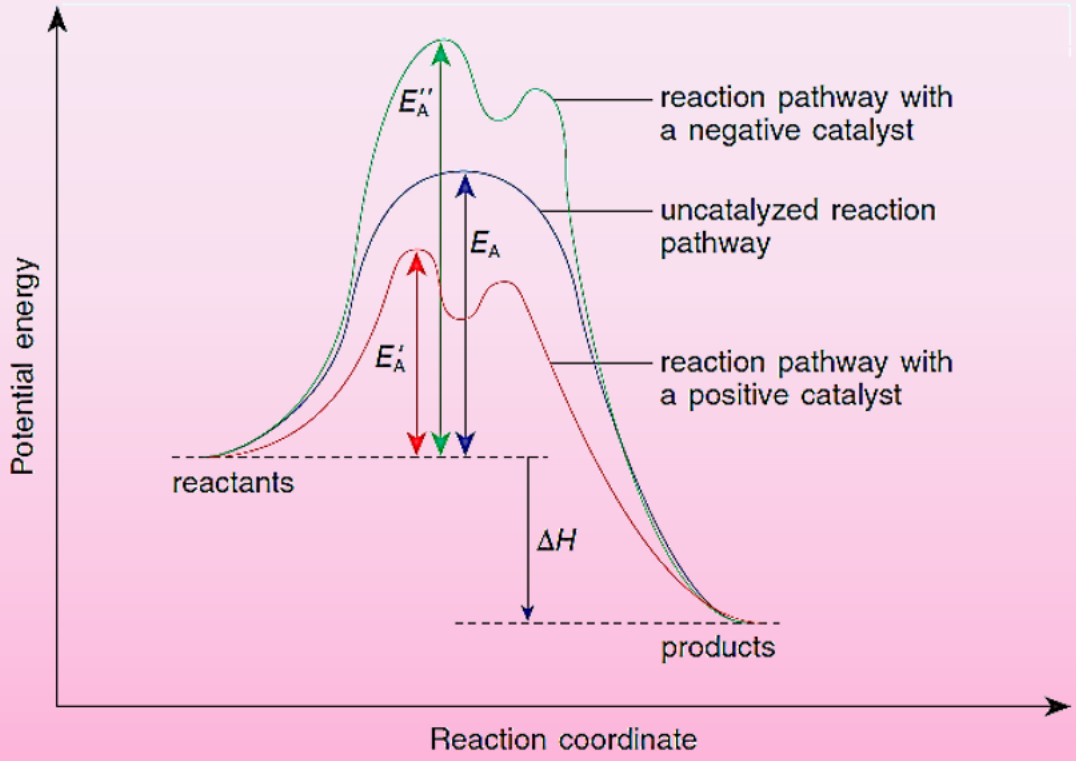
- If a reaction occurs in more than one step, it must involve species that are neither the reactant nor the final product
- These are called **reaction intermediates** or simply "intermediates"
- Each step has its own free energy of activation
- The complete diagram for the reaction shows the free energy changes associated with an intermediate





Describing a Reaction: Enzyme Catalysed Reaction (Revision)

- An energy diagram for a typical, **enzyme-catalyzed biological reaction** versus an **uncatalyzed reaction**.
- The biological reaction involves many steps, each of which has a relatively smaller activation energy and smaller energy change. The end result is the same.



Interpreting Organic Reactions Using ΔG

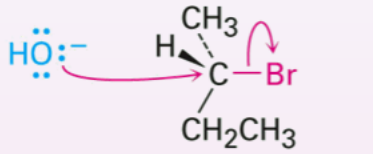
Example 1: Nucleophilic Substitution (S_N2)

❖ To favour reaction: **decreasing** $\Delta G^{\ddagger}$ by

(i) decreasing energy of TS (e.g. less hindered substrate or good leaving group)

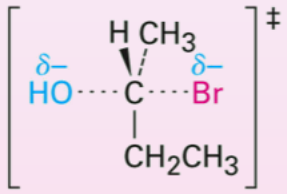
(ii) increasing ground state energy of reactants (e.g. good nucleophile and polar aprotic solvent)

(e.g. Substitution of 1° / 2° halide with OH^-, Substitution of 1° / 2° alcohol with $SOCl_2$ / PBr_3)



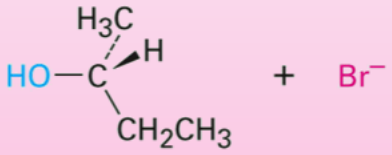
(*S*)-2-Bromobutane

1

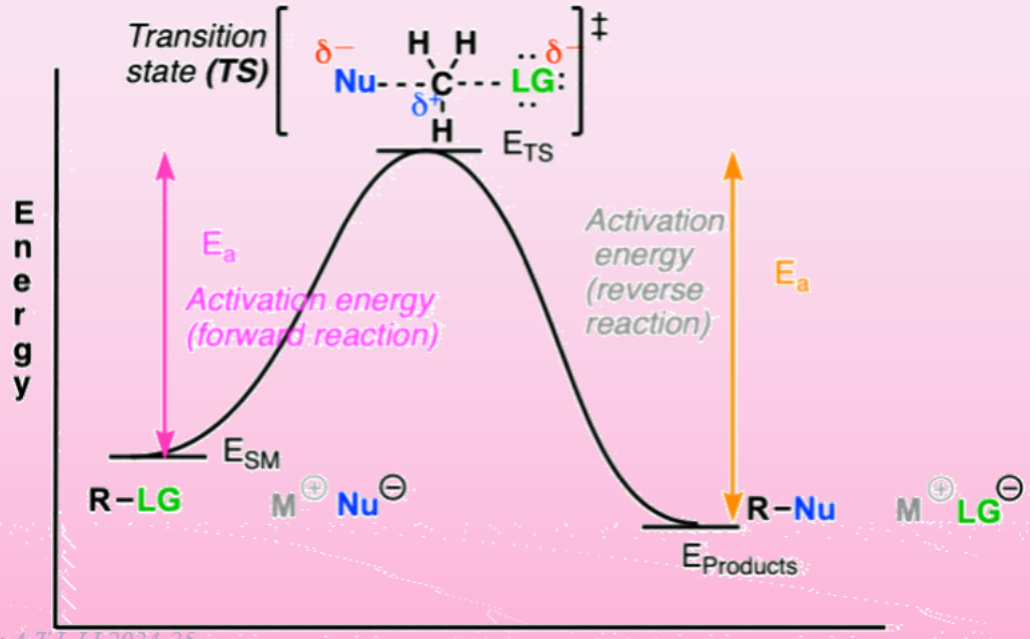


Transition state

2



(*R*)-2-Butanol



Interpreting Organic Reactions Using ΔG

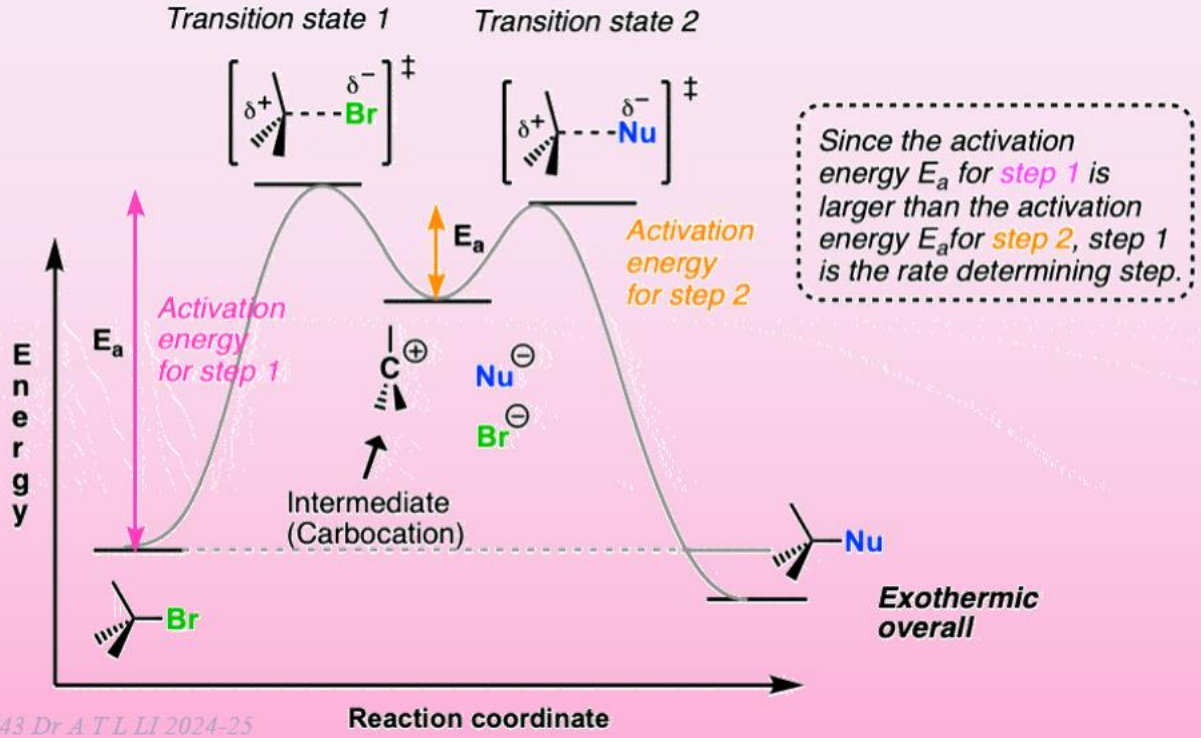
Example 2: Nucleophilic Substitution (S_N1)

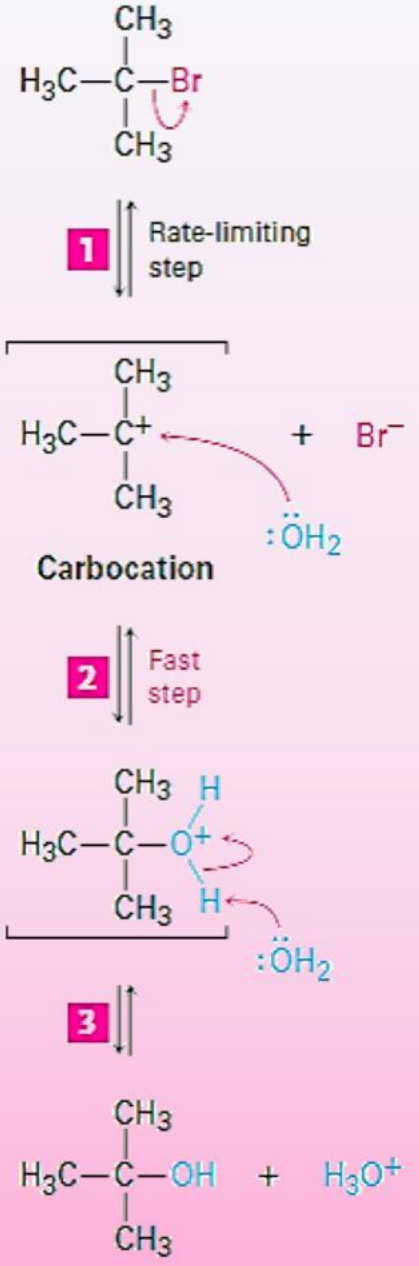
- To favour reaction:

(i) **decreasing** $\Delta G^{\ddagger}_1$ by decreasing energy of TS (e.g. good leaving group)

(ii) Stabilize carbocation intermediate by using 3° or allylic substrate, or polar protic solvent)

(e.g. Substitution of 3° halide with neutral nucleophile, Substitution of 3° alcohol with HCl / HBr)



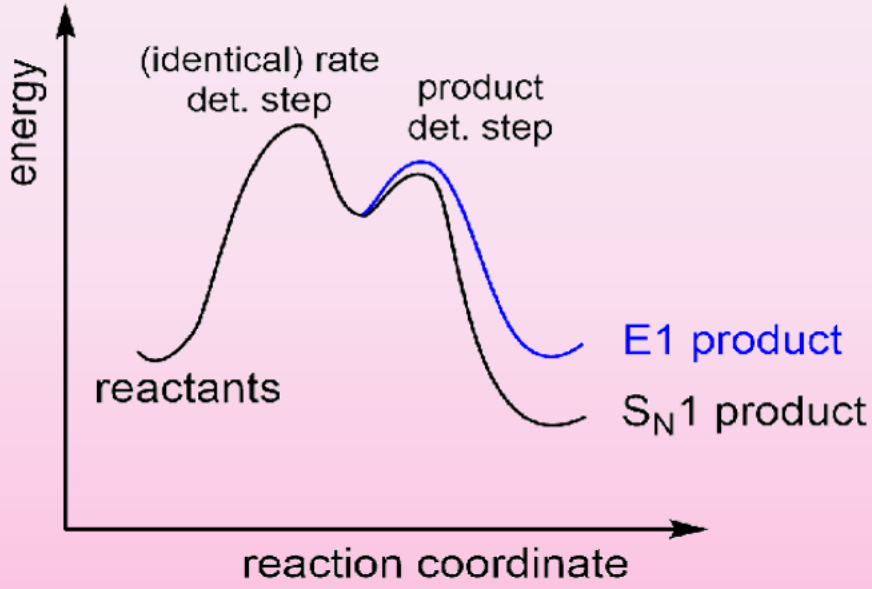


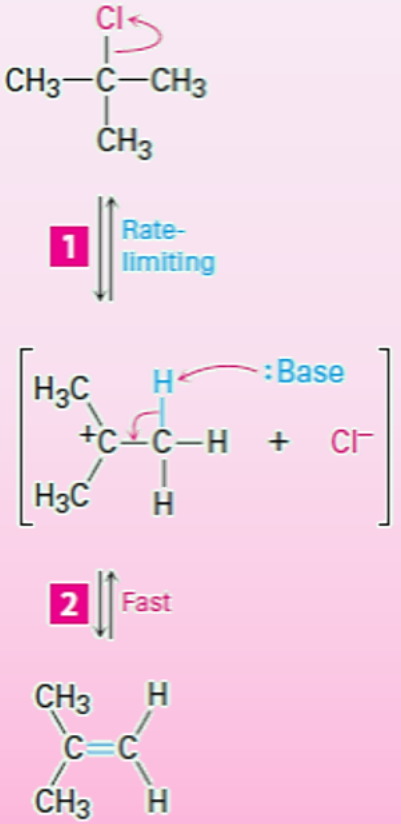
Interpreting Organic Reactions Using ΔG

Example 3: Elimination Reaction (*E*1)

- Same r.d.s. as S_N1 but different $\Delta G^{\ddagger}_2$
- Weak base and higher temperature favour *E*1

(e.g. Elimination of of 3° halide with weak base, Dehydration of 3° alcohol with H_3O^+ , with increasing conc and temperature for less substituted alcohol)



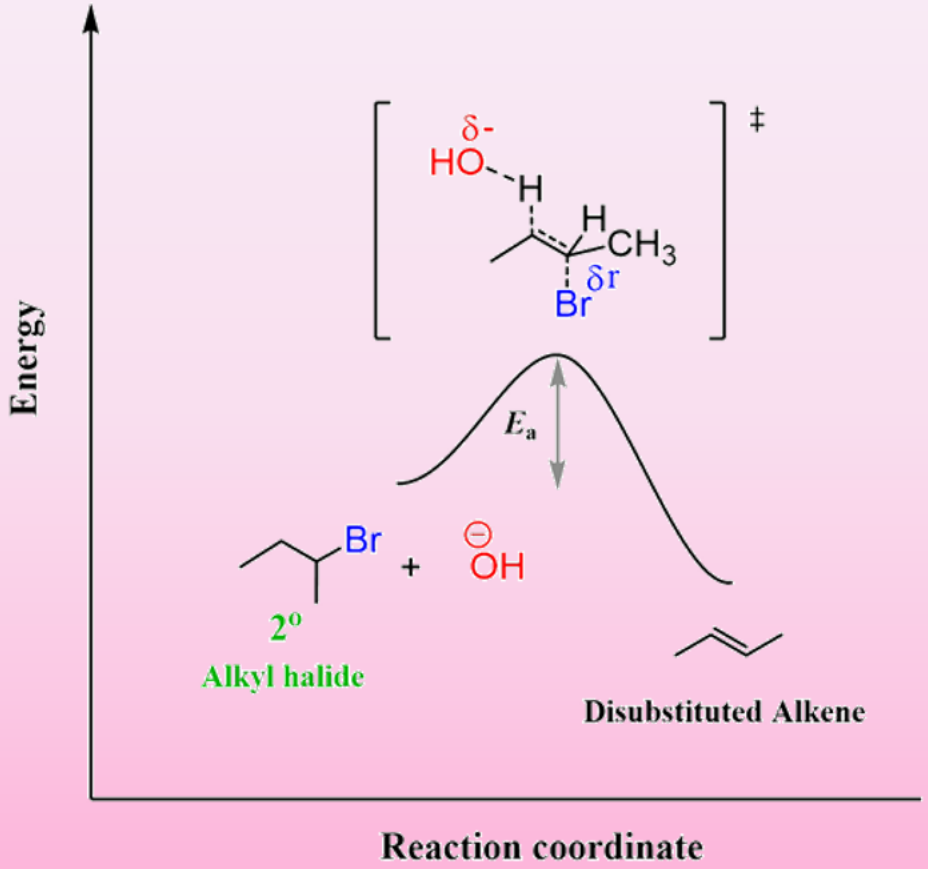


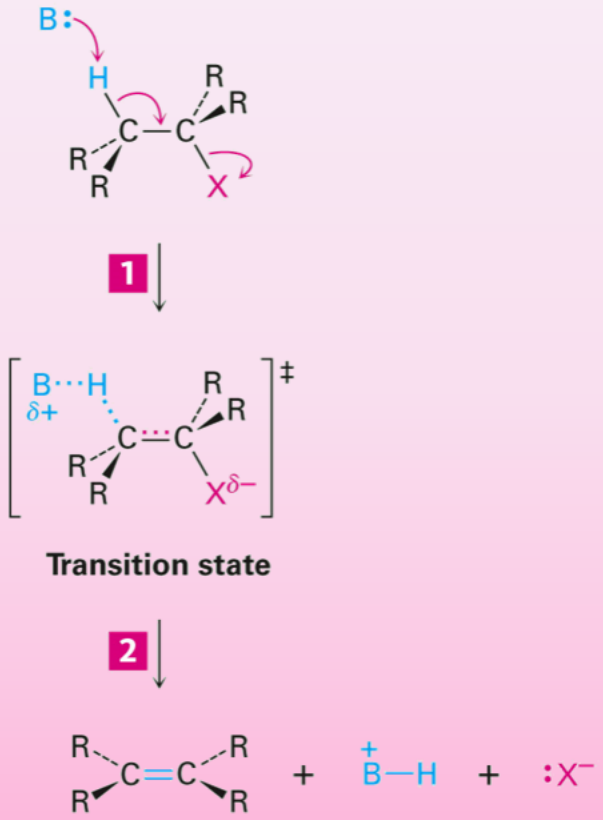
Interpreting Organic Reactions Using ΔG

Example 4: Elimination Reaction (*E*2)

- Relatively low $\Delta G^{\ddagger}$
- Strong base and higher concentration favour *E*2

(e.g. Elimination of of 3° halide with OH^- , and 1° halide with bulky alkoxides (e.g. $^-OC(CH_3)_3$))





Coupled Reactions

- Two ways to obtain product from a non-spontaneous reaction:
  (1) Change the reaction conditions to ones that make the reaction spontaneous (mostly by changing T), and
  (2) Combine a pair of reactions, one with a positive $\Delta_r G$, and one with a negative $\Delta_r G$ to obtain a spontaneous overall reaction. Such paired reactions are called **coupled reactions**.

Example: Considering the extraction of a metal from its oxide

- When copper(I) oxide is heated to 673 K, no copper metal is obtained. The decomposition of Cu_2O to form products in their standard states (for instance, P_{O2} = 1.00 bar) is nonspontaneous at 673 K.

$$Cu_2O(s) \xrightarrow{\Delta} 2\,Cu(s) + \frac{1}{2}\,O_2(g) \qquad \Delta_r G^\circ_{673\,K} = +125\ kJ\ mol^{-1}$$

Coupled Reactions

- Suppose this non-spontaneous decomposition reaction is coupled with the partial oxidation of carbon to carbon monoxide — a spontaneous reaction. The overall reaction is spontaneous when reactants and products are in their standard states because has $\Delta_r G^{\ominus}$ a negative value.

$$Cu_2O(s) \longrightarrow 2\,Cu(s) + \frac{1}{2}O_2(g) \qquad \Delta_r G^{\circ}_{673\,K} = +125\ kJ\ mol^{-1}$$

$$C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g) \qquad \Delta_r G^{\circ}_{673\,K} = -175\ kJ\ mol^{-1}$$

$$Cu_2O(s) + C(s) \longrightarrow 2\,Cu(s) + CO(g) \qquad \Delta_r G^{\circ}_{673\,K} = -50\ kJ\ mol^{-1}$$

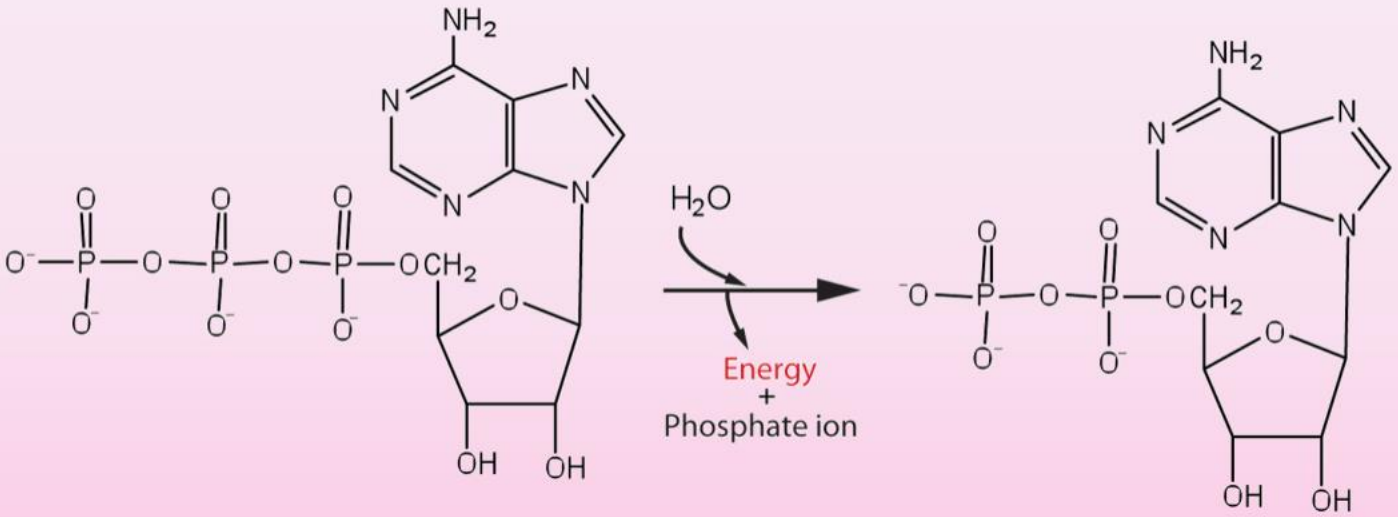
- The purpose of coupled reactions, then, is to produce a spontaneous overall reaction by combining two other processes: one non-spontaneous and one spontaneous.

- Many metallurgical processes employ coupled reactions, especially those that use carbon or hydrogen as reducing agents to extract metals. (e.g. C / CO reduction (reactivity $\leq$ Zn) to CO_2 (*g*) and H_2 reduction (reactivity $\leq$ Fe) to H_2O (*g*))

Coupled Reactions

- This is a common feature in biological systems where some enzyme-catalysed reactions are interpretable as two coupled half-reactions, one spontaneous and the other non-spontaneous.

- Organisms often the *hydrolysis* of **ATP** (*adenosine triphosphate*) to generate **ADP** (*adenosine diphosphate*) as the spontaneous coupling reaction.



ATP (*adenosine triphosphate*) **ADP** (*adenosine diphosphate*)

$$ATP + H_2O \rightleftharpoons ADP + P_i$$

(where P_i is inorganic phosphate ion)

$$\Delta_r G = -31 \text{ kJ/mol}$$

Coupled Reactions

- ATP is the major 'energy' molecule produced by metabolism, and it serves as a sort of 'energy source' in cell: ATP is dispatched to wherever a non-spontaneous reaction needs to occurs so that the two reactions are coupled so the overall reaction is thermodynamically favoured.

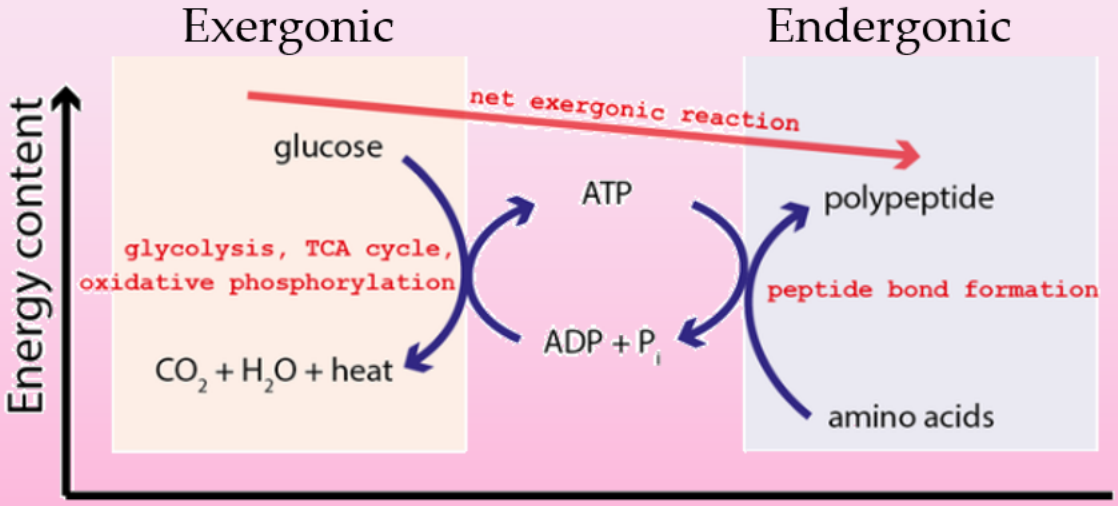
Example:

$ATP + H_2O \rightarrow ADP + P_i$ $\quad \Delta_r G$: – 31 kJ/mol

amino acids → polypeptides $\quad \Delta_r G$: ~ + 10 kJ/mol

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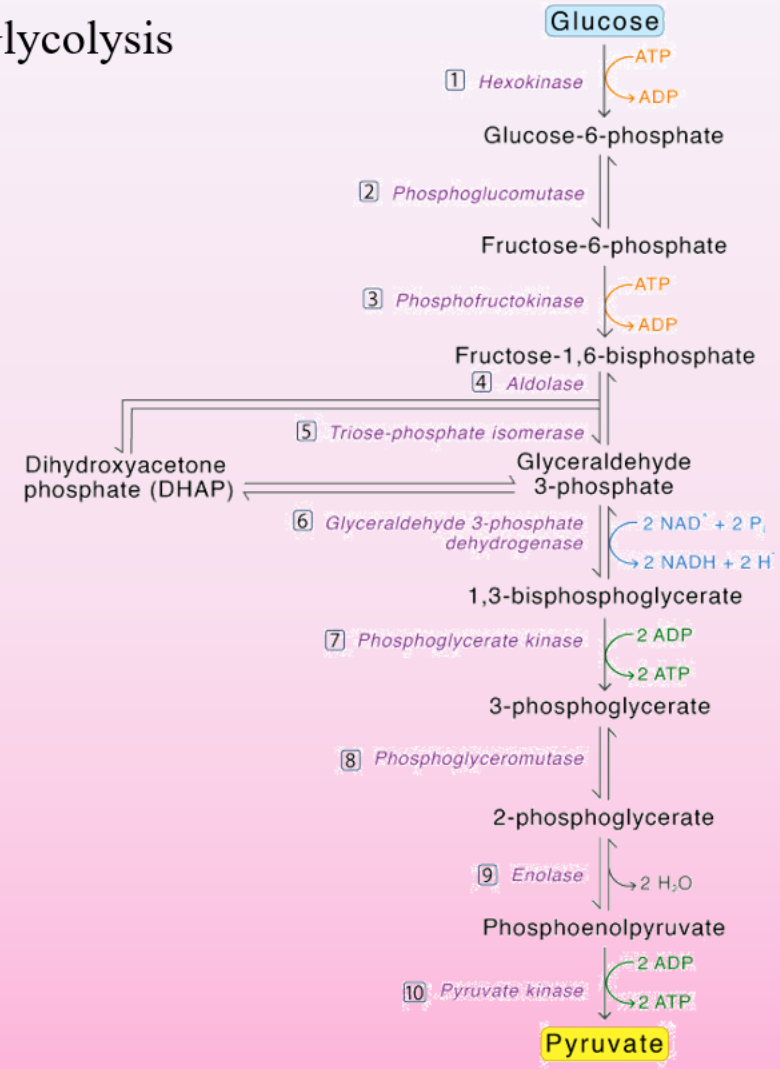
overall $\Delta_r G$: ~ – 21 kJ/mol



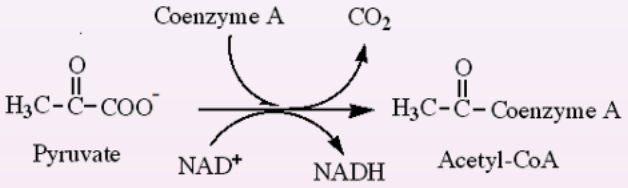
Coupled Reactions

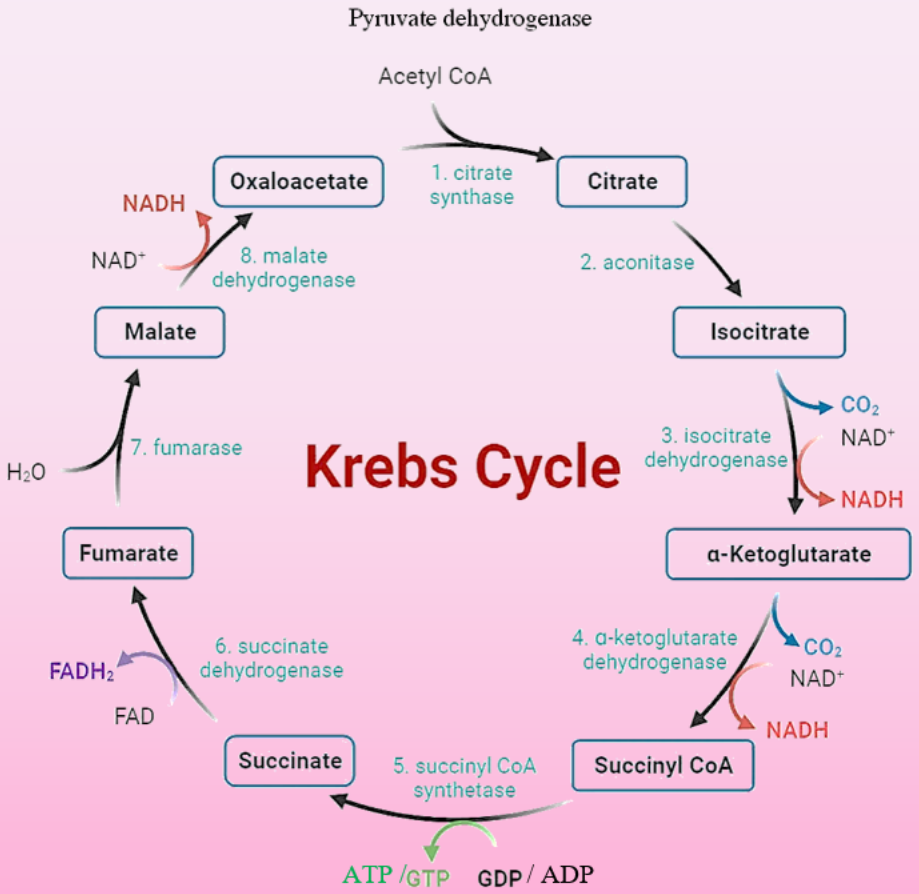
- Where does the "energy" come from?
- Cellular Respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ $\quad \Delta_r G = -2880$ kJ/mol

Glycolysis



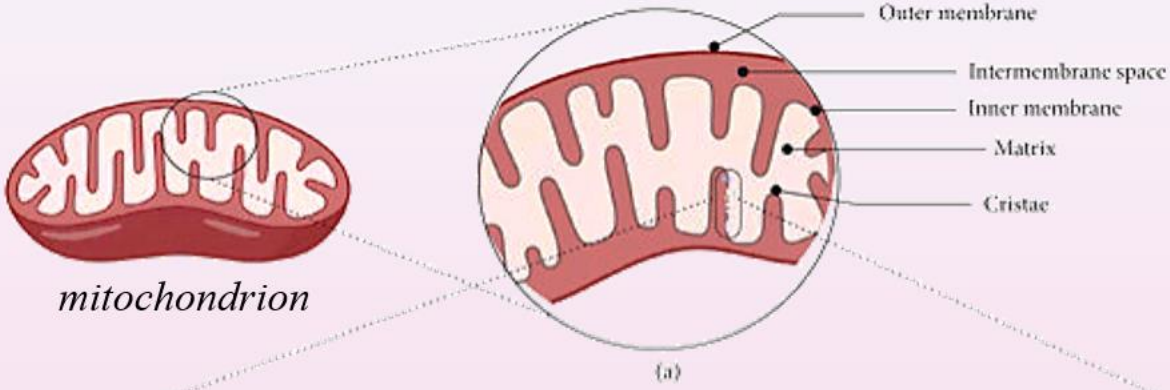
Linked Reaction



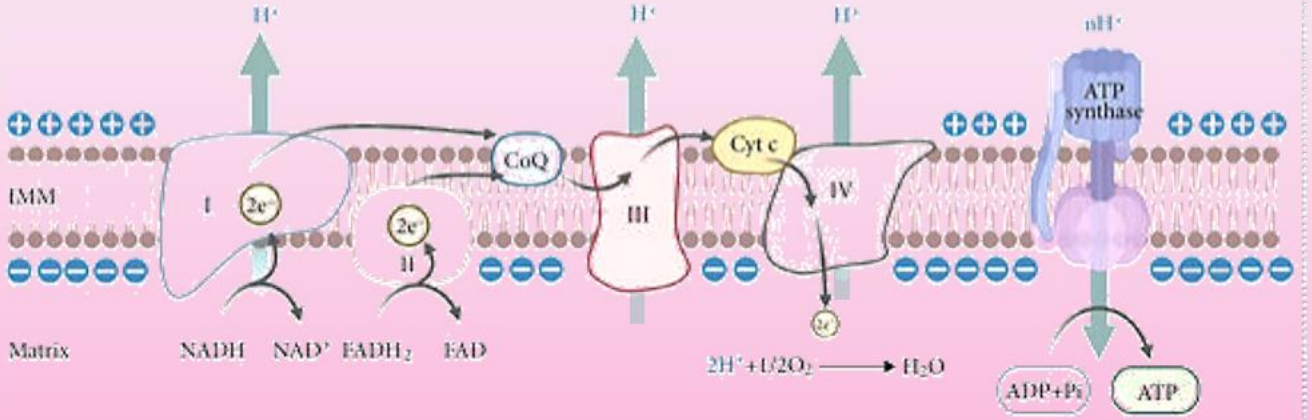


Coupled Reactions

- Oxidation Phosphorylation
- Each NADH (*reduced nicotinamide adenine dinucleotide*) could generate 3 ATPs
- Each $FADH_2$ (*reduced flavin adenine dinucleotide*) could generate 2 ATPs



Electron Transport Chain (ETC)



- How many ATPs are produced during respiration of 1 glucose molecule?
- Could you find out the coupled reactions involved in the overall respiration process?

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